Lab Assignment 7: Routing Protocols Simulation in Python

October 8, 2026

Objective

The objective of this lab assignment is to simulate the working of various routing protocols in Python. Students are expected to understand the key concepts, algorithms, and behaviors of each protocol and demonstrate them using simulation-based programs.

- RIP (Routing Information Protocol)
- OSPF (Open Shortest Path First)
- BGP (Border Gateway Protocol)
- IS-IS (Intermediate System to Intermediate System)

General Instructions

1. Each part should be implemented as a separate Python file.
2. Use graphs or suitable data structures to represent network topology.
3. Provide clear documentation and comments in your code.
4. Include sample outputs and screenshots of routing tables or path trees.
5. Submit a single ZIP file containing:
    - Source code files
    - Output screenshots
    - A brief report (PDF) summarizing your observations and results

Part 1: RIP (Routing Information Protocol)

Task

Simulate the RIP protocol using the Bellman–Ford algorithm. Each router should maintain a distance vector containing the number of hops to each destination. Routers should periodically exchange their routing tables until convergence.

Steps

1. Create a network topology (graph) with routers as nodes and links as edges.
2. Implement the Bellman–Ford algorithm to calculate the shortest path.
3. Simulate periodic routing updates between routers.
4. Display each router's routing table after convergence.
5. Comment on convergence behavior and message overhead.

Part 2: OSPF (Open Shortest Path First)

Task

Simulate the OSPF protocol using Dijkstra's algorithm. Each router should maintain a link-state database and exchange link-state advertisements (LSAs) to synchronize the network topology.

Steps

1. Create a network graph with link costs representing bandwidth or latency.
2. Implement Dijkstra's algorithm to compute the shortest path tree for each router.
3. Simulate the exchange of LSAs between routers.
4. Display the shortest path tree and routing table for each router.
5. Discuss the differences in convergence and scalability compared to RIP.

Part 3: BGP (Border Gateway Protocol)

Task

Simulate BGP using a simplified path-vector protocol. Each router (representing an Autonomous System) should exchange AS paths and select the best route based on the shortest AS path.

Steps

1. Define an AS-level topology with routers as nodes and AS relationships as edges.
2. Implement path selection based on AS path length.
3. Simulate BGP UPDATE message exchanges between neighboring ASes.
4. Ensure loop prevention by discarding routes containing the local AS number.
5. Show final routing tables for all routers after convergence.

Part 4: IS-IS (Intermediate System to Intermediate System)

Task

Simulate IS-IS using a link-state routing approach similar to OSPF. Each router should build a complete view of the network topology and compute the shortest paths to all destinations.

Steps

1. Create a graph representing routers and link metrics.
2. Implement link-state flooding and database synchronization.
3. Use Dijkstra's algorithm to compute shortest paths.
4. Display the final routing table for each router.
5. Compare IS-IS with OSPF in terms of functionality and deployment.

Deliverables

1. Source code files for each protocol simulation.
2. A short report (2–3 pages) including:
    - Network topology diagram
    - Routing table snapshots
    - Comparison of convergence behavior and overhead
    - Observations and conclusions
3. Screenshots of program output.

Additional Notes

- You may use Python libraries such as `networkx` for graph representation.
- Do not use pre-built routing libraries.
- Bonus marks for visualizing network convergence or message propagation.